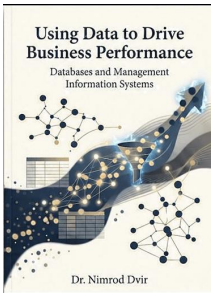
# Using Data to Drive Business Performance




**Databases and Management Information Systems**
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## A business question. A reliable answer. A better decision.

Courseware that connects data and business performance: a textbook, interactive reader, guided builds, independent labs, reading assessments and a semester project. Students organize data, query it with SQL, check the results, present the evidence and recommend an action.

### Who it is for

**Primary:** the applied business database or MIS data-management course. **A second route:** the business-core data or IS course. Selected modules can also support accounting information systems and analytics foundations.

### Why I built it

After seven years of teaching this course to more than 4,000 students, I still found myself combining books and writing the missing pieces. I built the courseware I wanted to teach. Version 1 was tested with a small group in summer 2026; version 2 is in use with 500 students in fall 2026.

### What makes the proposed edition different

**Begin with the business:** performance, goals and KPIs, followed by data literacy and information organization. **Use before design:** query a supplied database, then model and build one. **Carry the work through:** analytics, visualization and Power BI lead to a decision memo. **Make verification a habit:** students explain and check their own and AI-assisted work.

### A substantial research foundation

**63 documents**
31 institutions

**60 titles**
Plus four detailed comparisons

**52.9%**
SQL in 855 relevant U.S. analyst postings

The market evidence includes **375,418 U.S. business bachelor's degrees** in 2021-22 (NCES) and a 2017 study in which **52 of 61 schools (85%)** included IS I in the business core. These measure audience scale and institutional course presence, not this book's enrollments. SQL figure: 365 Data Science, 2026; 855 relevant postings from 1,000 Indeed U.S. listings.

The syllabus and textbook audits inform the proposed sequence, platform routes, practice and project. Full methods, findings and source references are included in the research package.

# The proposed edition

**A discussion draft:** I would like to work with editors and reviewers to finalize the structure, tools, depth and running cases. The following is a research-informed starting point, not a settled manuscript.

**Part I · Chapters 1 to 3 · From Data to Decisions**
Chapter 1: From Data to Business Performance
Chapter 2: Understanding Data
Chapter 3: Organizing Data and Information

**Part II · Chapters 4 to 6 · Databases and SQL**
Chapter 4: Databases and the Relational Model
Chapter 5: SQL Foundations
Chapter 6: SQL for Business Analysis
Midterm studio: Integration Studio A (midterm)

**Part III · Chapters 7 to 11 · Designing and Running Databases**
Chapter 7: Database Design: From Requirements to ER Diagrams
Chapter 8: Normalization
Chapter 9: Building and Testing a Database
Chapter 10: Advanced SQL
Chapter 11: Database Administration, Security and Reliability

**Part IV · Chapters 12 to 15 · From Data to Insight**
Chapter 12: Business Analytics and Decision-Making
Chapter 13: Reporting and Data Visualization
Chapter 14: Power BI
Chapter 15: Big Data, Data Warehouses and AI

**Part V · Chapter 16 and Closing · Putting It to Work**
Chapter 16: Capstone: From Data to Decisions
Closing chapter: Closing: Where Data Can Take You

**Companions and appendices**
Supabase and the in-book editor; MySQL route; Microsoft Access route; SQL reference and dialect guide; diagramming; Tableau; Terms Treasury.

## Read, build, try, reflect

Every chapter includes an introduction, core concepts, a guided **Let's Build**, review and reflection questions, a **Reading Assessment Test**, an independent **interactive lab**, terms and takeaways. A preface sets up the course; a midterm studio and three project checkpoints lead to the capstone. A closing chapter connects the work to careers.

## Available today, and the work ahead

**In use:** interactive textbook, chapter builds, labs and quizzes, instructor materials and dashboard, Grading Database and PetVax cases, and Access labs and appendix. **Next edition:** the 16-chapter structure, planned PGlite browser practice and hosted Supabase project, Power BI with a Tableau alternative, and a MySQL route to develop and test. Automated SQL/lab feedback is a prototype in testing.

## Three decisions to make together

**1.** Publishing arrangement and portfolio positioning. **2.** Development scope, tools, cases and a sample unit. **3.** Responsibilities, review milestones and the manuscript submission timeline.

**Let's cement the publishing relationship.**
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Explore the live reader · Full research reports and detailed outline accompany this brief.